

Experiment No: 7

Title:

Perform DOM Traversal and Updates by Developing a To-Do List Application with Add, Edit, and Delete Features.

Software/Tools Required:

- Visual Studio Code
- Web Browser such as Google Chrome
- HTML, CSS, and JavaScript

Theory:

The **Document Object Model (DOM)** is a programming interface that represents an HTML document as a tree-like structure. Each HTML element is treated as an object called a **node**, allowing JavaScript to access, modify, add, or remove elements dynamically without reloading the webpage. DOM manipulation is one of the most important concepts in JavaScript for creating interactive web applications.

DOM traversal refers to navigating through the DOM tree to access parent, child, or sibling elements. JavaScript provides several properties and methods such as `parentElement`, `children`, `firstElementChild`, `lastElementChild`, `nextElementSibling`, `previousElementSibling`, `querySelector()`, and `querySelectorAll()` for traversing and selecting elements.

DOM updates involve modifying the content, attributes, styles, or structure of HTML elements. Methods such as `createElement()`, `appendChild()`, `removeChild()`, `replaceChild()`, `setAttribute()`, `innerHTML`, `textContent`, and `classList` are commonly used to dynamically update the webpage based on user interactions.

This experiment demonstrates how DOM traversal, element creation, modification, and deletion can be used to build a practical and interactive web application. It also highlights the importance of event-driven programming and dynamic content management in modern web development.

Result/Conclusion:

Thus, DOM traversal and DOM manipulation techniques were successfully implemented using JavaScript. A To-Do List application was developed that dynamically adds, edits, and deletes tasks, demonstrating the effective use of DOM methods, event handling, and real-time webpage updates.

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